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Inkjet recording apparatus and head position determining method

Abstract

An inkjet recording apparatus captures an image of a recording surface of a recording medium including a test pattern recorded on the recording surface, calculates an interval between the test pattern and a reference position on the recording surface based on the image of the recording surface, and determines a height position of the recording head according to the recording medium based on the interval. The inkjet recording apparatus outputs information related to the determined height position.

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2022-042313 filed on Mar. 17, 2022, the entire contents of which are incorporated herein by reference.

(2) The present disclosure relates to an inkjet recording apparatus that records an image on a recording medium by ejecting ink from a recording head onto the recording medium, and a head position determining method that determines a position of the recording head.

BACKGROUND

(3) Generally, an inkjet recording apparatus that records an image on a recording medium such as fabric or paper by ejecting ink from nozzles of a recording head is known. Printing an image on a fabric such as cloth is called textile printing, and an inkjet recording apparatus that performs textile printing is called a textile printing apparatus or a textile printer. In the inkjet recording apparatus, when a thick recording medium is set in the apparatus and printed, a nozzle surface of the recording head may come into contact with a surface of the recording medium. Therefore, as a conventional apparatus, a recording apparatus capable of determining a distance between the nozzle surface and the recording medium is known.

SUMMARY

(4) An inkjet recording apparatus according to one aspect of the present disclosure includes a recording head, a pattern print processing portion, an imaging portion, a calculation processing portion, and a determination processing portion. In addition, the inkjet recording apparatus includes either one of an output processing portion, or a head position adjustment portion and a position adjustment processing portion. The recording head ejects ink onto a recording medium while moving in a predetermined scanning direction. The pattern print processing portion causes the recording head to record a specified test pattern on a recording surface of the recording medium.

The imaging portion captures an image of the recording surface including the test pattern. The calculation processing portion calculates an interval between the test pattern and a reference position on the recording surface based on the image of the recording surface. The determination processing portion determines a height position of the recording head according to the recording medium based on the interval.

(5) The output processing portion outputs information related to the height position determined by the determination processing portion.

(6) The head position adjustment portion is configured to be able to adjust the height position by changing a relative position of the recording head with respect to a support surface of the recording medium. The position adjustment processing portion causes the head position adjustment portion to change the relative position of the recording head based on the height position determined by the determination processing portion.

(7) A head position determining method according to another aspect of the present disclosure includes a test pattern printing step, an imaging step, an interval calculation step, and a height position determination step. The test pattern printing step records a specified test pattern on a recording surface of a recording medium by ejecting ink from a recording head while moving the recording head in a predetermined scanning direction. The imaging step captures an image of the recording surface including the test pattern. The interval calculation step calculates an interval between the test pattern and a reference position on the recording surface based on the image of the recording surface. The height position determination step determines a height position of the recording head according to the recording medium based on the interval.

(8) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description with reference where appropriate to the accompanying drawings. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram showing a structure of an inkjet recording apparatus of an embodiment according to the present disclosure.

(2) FIG. 2 is a view as seen from above a conveying unit and a recording unit of the inkjet recording apparatus shown in FIG. 1.

(3) FIG. 3 is a view, as seen from a downstream side in the conveying direction of a recording medium, of the conveying unit and the recording unit of the inkjet recording apparatus shown in FIG. 1.

(4) FIG. 4A is a schematic diagram for explaining a cause of deterioration of image quality.

(5) FIG. 4B is a schematic diagram for explaining a cause of deterioration of image quality.

(6) FIG. 5 is a block diagram showing a configuration of an inkjet recording apparatus.

(7) FIG. 6 is a diagram showing each test pattern printed on a recording surface of a recording medium.

(8) FIG. 7 is a flowchart showing an example of a procedure of a head height adjustment process executed by a control portion of an inkjet recording apparatus.

DETAILED DESCRIPTION

(9) Embodiments according to the present disclosure will be described below with reference to the drawings. Note that the embodiments described below are examples of implementing techniques according to the present disclosure and do not limit the technical scope of the present disclosure.

(10) An inkjet recording apparatus **10** (hereinafter abbreviated as “recording apparatus **10**”) of an embodiment according to the present disclosure will be described with reference to FIGS. **1** to **3**. FIG. **1** is a diagram showing a structure of the recording apparatus **10**. FIG. **2** is a view as seen from above of a conveying unit **12** and a recording unit **14** of the recording apparatus **10**. FIG. **3** is a view of the conveying unit **12** and the recording unit **14** of the recording apparatus **10** as seen from a downstream side in a conveying direction **D1** of a recording medium **Ct**.

(11) The recording apparatus **10** is an apparatus that, based on input image data, records an image on a sheet-like recording medium **Ct** such as a non-woven fabric or cloth by ejecting ink onto an upper surface (recording surface) of the recording medium **Ct** (see FIG. **3**) and dyeing the fibers of the recording medium **Ct** with ink. In this embodiment, the recording apparatus **10** will be described as an inkjet recording apparatus mainly used for textile printing. That is, the recording apparatus **10** is a so-called textile printing machine (textile printer). Here, the recording medium **Ct** is merely an example of a recording medium according to the present disclosure, and is a sheet-like fabric such as non-woven fabric or cloth. Note that, in this embodiment, for example, various types of paper having different thicknesses may be applied as other examples of recording media on which images are recorded.

(12) As shown in FIG. **1**, the recording apparatus **10** includes a feed unit **11**, a conveying unit **12**, a discharge unit **13**, a recording unit **14**, a raising/lowering mechanism **15** (an example of a head position adjustment portion according to the present disclosure), a maintenance unit **16**, a cap unit **17**, an operation display portion **18** (an example of a specified display portion according to the present disclosure), an imaging portion **19** (an example of an imaging portion according to the present disclosure), a control device **80**, and a body frame **21** that accommodates or supports these units, and the like.

(13) The feed unit **11** has a driving roller **11A**, a driven roller **11B**, and an annular belt **11C** stretched over the driving roller **11A** and the driven roller **11B**. The driving roller **11A** and the driven roller **11B** are horizontally spaced apart, and thus the annular belt **11C** extends horizontally. A recording medium **Ct** on which an image is to be recorded is set on an upper surface of the annular belt **11C**. The drive roller **11A** is connected to a motor (not shown) via a transmission member such as a gear or belt. The drive roller **11A** is rotationally driven in the counterclockwise direction in FIG. **1** by driving and controlling the motor by a control signal from a control device **80** of the recording apparatus **10**.

(14) Thus, the recording medium **Ct** held on the upper surface of the annular belt **11C** is conveyed in a conveying direction **D1**.

(15) The conveying unit **12** is provided further on the downstream side than the feed unit **11** in the conveying direction **D1**. The conveying unit **12** supports the recording medium **Ct** and conveys the recording medium **Ct** toward the downstream side in the conveying direction **D1**. That is, the conveying unit **12** functions as a support portion that supports the recording medium **Ct**, and also functions as a conveying portion that conveys the recording medium **Ct** in the conveying direction **D1**.

(16) The conveying unit **12** has a driving roller **12A**, a driven roller **12B**, and an annular belt **12C** stretched over the driving roller **12A** and the driven roller **12B**. The conveying unit **12** is arranged at the same position in the height direction as the feed unit **11** so as to be horizontal with the feed unit **11**. In addition, in the conveying unit **12** as well, the drive roller **12A** is connected to a motor (not shown) via a transmission member such as a gear or belt. The drive roller **12A** is rotationally driven by the motor being driven and controlled by a control signal from the control device **80**. Thus, the recording medium **Ct** supported by a conveying surface **22** (an example of a supporting surface according to the present disclosure) of the annular belt **12C** is conveyed in the conveying direction **D1**. Here, the conveying surface **22** is an upper surface of an upper portion (hereinafter referred to as “belt upper portion”) **12C1** of the annular belt **12C**.

(17) In the present embodiment, when a recording head **26** (an example of a recording head

according to the present disclosure) of the recording unit **14** conveys a predetermined recording area on an upper recording surface of the recording medium Ct to a recording position (position below the recording head **26**), conveying the recording medium Ct is paused. After that, the recording unit **14** is moved back and forth in a scanning direction D2 (see FIG. 2) from a specified standby position. Note that the scanning direction D2 is a direction perpendicular to the conveying direction D1 of the recording medium Ct.

(18) While the recording unit **14** moves back and forth in the scanning direction D2, the recording head **26** ejects ink from nozzles **29** (see FIG. 2) toward the recording area. Thus, an image is recorded with ink in the recording area. After that, the recording medium Ct is conveyed again in the conveying direction D1 in order to convey the next recording area on the recording medium Ct to the recording position. When the next recording area reaches the recording position, conveying is paused again, and the ink ejection operation by the recording head **26** is performed. That is, the recording medium Ct is intermittently conveyed in the conveying direction D1 each time image recording is performed by the recording head **26**. A specified image is recorded on the recording surface of the recording medium Ct by repeating the intermittent conveying operation of the recording medium Ct and the ejection operation of the ink by the recording head **26**.

(19) The recording unit **14** records an image on the recording surface of the recording medium Ct by ejecting ink onto the recording surface of the recording medium Ct while moving back and forth in the scanning direction D2. The recording unit **14** is supported at a position a specified interval above the conveying unit **12** so as to be able to move back and forth in the scanning direction D2.

(20) The recording unit **14** includes a carriage **141**, four recording heads **26** held by the carriage **141**, and a motor **25** for applying a driving force to the carriage **141** to move the carriage **141** back and forth in the scanning direction D2. The carriage **141** is supported by two guide rails **31** (see FIG. 2) provided on the body frame **21** so as to be able to freely move in the scanning direction D2. The carriage **141** is connected to the motor **25** (see FIG. 5) via a transmission member such as gear or belt. The recording unit **14** moves back and forth in the scanning direction D2 by the motor **25** being driven and controlled in the forward rotation direction or the reverse rotation direction by a control signal from the control device **80**.

(21) The recording heads **26** are respectively connected to corresponding ink tanks (not shown) by ink tubes (not shown). Inks of four colors (cyan, magenta, yellow, and black) stored in the ink tanks are supplied to the recording heads **26** of the corresponding colors through the ink tubes. Note that the recording heads **26** in this embodiment are an example of a recording head according to the present disclosure.

(22) A lower surface of each recording head **26** is an ink ejection surface **28** on which a plurality of nozzles **29** are formed. That is, the lower surface of the recording head **26** is provided with an ink ejection surface **28** on which a plurality of nozzles **29** are formed. A plurality of rows of nozzles **29** are formed along the conveying direction D1. Each nozzle **29** has a lower opening. The lower opening is an ink ejection port for ejecting ink droplets onto the surface of the recording medium Ct.

(23) The recording head **26**, by being controlled by a control signal from the control device **80** (see FIG. 1), ejects ink droplets from the nozzles **29** toward the recording surface of the recording medium Ct according to image data received from an external computer. Thus, a color image is recorded on the recording medium Ct on the annular belt **12C** with four colors of ink, cyan, magenta, yellow, and black.

(24) Note that the recording unit **14** is not limited to a configuration having a plurality of recording heads **26** corresponding to each color, and, for example, may be configured to have a plurality of recording heads **26** each having the same color, or may be configured to have one recording head **26** to record a monochromatic image on the recording medium Ct.

(25) As shown in FIG. 3, an interval T is provided in the height direction between the ink ejection surface **28** of the recording head **26** and the conveying surface **22** of the annular belt **12C**. That is,

the head height H of the ink ejection surface **28** of the recording head **26** from the conveying surface **22** (the height position of the recording head **26**) is determined so that the interval T is formed between the ink ejection surface **28** and the conveying surface **22**. In other words, the head height H is the distance in the height direction from the conveying surface **22** to the ink ejection surface **28**.

(26) The interval T is an interval that allows the recording medium Ct to be conveyed in the conveying direction D1 without the recording surface of the recording medium Ct on the conveying surface **22** coming into contact with the ink ejection surface **28**, and, for example, the interval T is a distance obtained by adding a specified design interval t0 (for example, 1 mm) to the thickness of the recording medium Ct. The design interval t0 is an optimum value for ensuring image quality equal to or higher than a predetermined reference level. That is, the optimum head height H is a distance obtained by adding the design interval t0 to the thickness of the recording medium Ct, and an optimum height position of the recording head **26** is a position above the conveying surface **22** by the distance.

(27) In a case where the thickness of the recording medium Ct on which an image is to be recorded changes, a separation distance t1 (see FIG. 3) between the ink ejection surface **28** and the recording surface also changes, and the separation distance t1 deviates from the designed interval t0. That is, the head height H deviates from the optimum value that can ensure image quality equal to or higher than the reference level. In this case, the quality of the recorded image is degraded.

(28) For example, as shown in FIG. 4A, when the thickness of the recording medium Ct is too small with respect to the interval T, the separation distance t1 becomes larger than the design interval t0. In this case, ink droplets are ejected during scanning by the recording unit **14**, and the distance required for the ink droplets to reach the recording surface increases. Therefore, in a case where the recording unit **14** moves in one moving direction D21 of the scanning direction D2, the ink droplet landing position P1 shifts a distance Δd in the moving direction D21 from a reference landing position P0 at the design interval t0. Such a deviation (ad) similarly occurs in a case where the recording unit **14** moves in an other movement direction D22. In this case, the recorded image becomes blurred or unclear, and the image quality deteriorates.

(29) In addition, for example, as shown in FIG. 4B, when the thickness of the recording medium Ct is too large with respect to the interval T, the separation distance t1 becomes smaller than the design interval t0. In this case, the interval between the ink ejection surface **28** and the recording surface of the recording medium Ct becomes too small, and the ink ejection surface **28** may come into contact with the recording surface. In this case, image rubbing occurs due to contact, and image quality deteriorates.

(30) Therefore, the recording apparatus **10** of the present embodiment is provided with a raising/lowering mechanism **15**, and as will be described later, is configured so that the distance of the interval T, that is, the relative position of the recording head **26** with respect to the recording medium Ct can be adjusted by the raising/lowering mechanism **15**.

(31) As shown in FIG. 1, an imaging portion **19** is provided above the conveying surface **22** of the annular belt **12C**. More specifically, the imaging portion **19** is provided above the conveying surface **22** and further on the downstream side than the recording unit **14** in the conveying direction D1. The imaging portion **19** is a device that reads an image on the recording surface of the recording medium Ct on the conveying surface **22**, and is, for example, a contact image sensor (CIS) having a plurality of imaging elements arranged in the scanning direction D2. In the imaging portion **19**, an imaging element receives light emitted from a light source onto the recording surface of the recording medium Ct and reflected from the recording surface, and the imaging portion **19** generates image data having a density corresponding to intensity of the light.

(32) In this embodiment, the imaging portion **19** reads an image in an entire width direction on the recording surface of the recording medium Ct that is moved in the conveying direction D1. In a case where a test pattern, which will be described later, is recorded on the recording surface of the

recording medium Ct by the recording head 26, the imaging portion 19 reads an image on the recording surface including the test pattern based on a control signal from the control device 80. Data of the read image is stored in a RAM 83 (see FIG. 5), a storage device 84 (see FIG. 5), or the like by the control device 80.

(33) As shown in FIG. 1, the discharge unit 13 is provided further on the downstream side than the conveying unit 12 in the conveying direction D1. The discharge unit 13 has a drive roller 13A, a driven roller 13B, and an annular belt 13C stretched over the drive roller 13A and the driven roller 13B. In the discharge unit 13 as well, the drive roller 13A is connected to a motor (not shown) via a transmission member such as a gear or belt. The drive roller 13A is rotationally driven by the motor being driven and controlled by a control signal from the control device 80. Thus, the recording medium Ct held on the upper surface of the annular belt 13C is conveyed in the conveying direction D1.

(34) In this embodiment, in a case where the recording medium Ct is at a position where recording by the recording head 26 of the recording unit 14 is possible, the discharge unit 13 is intermittently driven in synchronization with the conveying unit 12. When the recording medium Ct is completely transferred to the discharge unit 13, the discharge unit 13 is continuously driven to eject the recording medium Ct from the discharge port 24 to the outside. A discharge tray 23 is provided outside the discharge port 24, and the discharged recording medium Ct is held by the discharge tray 23. Note that ink adhering to the recording medium Ct naturally dries while passing through the discharge unit 13.

(35) The raising/lowering mechanism 15 is provided below the conveying unit 12. The raising/lowering mechanism 15 adjusts the head height H (distance of the interval T) of the recording head 26 by raising or lowering the conveying unit 12 and changing the relative position of the recording head 26 with respect to the conveying surface 22 of the recording medium Ct. The raising/lowering mechanism 15 includes a lift plate 15A that supports the conveying unit 12, a link arm 15B attached to the lift plate 15A, a main portion 15C that supports the link arm 15B so as to be vertically movable, and a motor 32 that applies a driving force to the link arm 15B. Note that the raising/lowering mechanism 15 is not limited to such a configuration, and any configuration can be applied as long as the mechanism is capable of raising and lowering the conveying unit 12 by the driving force of the motor 32.

(36) The motor 32 is provided with a rotational position sensor such as a rotary encoder. The control portion 81 can control the raised or lowered position of the lift plate 15A and the amount of raising or lowering of the lift plate 15A based on an output value of the rotary encoder. In addition, the control portion 81 can also control a relative position of the recording head 26 based on the output value of the rotary encoder. Further, in a case where a user inputs a setting value related to the head height H from an operation display portion 18, the control portion 81 can drive and control the raising/lowering mechanism 15 so that the position becomes the position indicated by the setting value.

(37) The raising/lowering mechanism 15 supports the conveying unit 12 from below and vertically raises or lowers the conveying unit 12 with respect to the recording head 26. That is, the raising/lowering mechanism 15 separates or brings closer together the conveying unit 12 and the recording head 26 by moving the conveying unit 12 relative to the recording head 26. More specifically, the raising/lowering mechanism 15 moves the conveying unit 12 between a recording position (position indicated by a solid line in FIG. 1) where printing by the recording unit 14 is possible, and a maintenance position (position indicated by a dashed line in FIG. 1) a specified distance below the recording position.

(38) In addition, in the raising/lowering mechanism 15, the motor 32 is driven and controlled by a control signal from the control device 80. Thus, the raising/lowering mechanism 15 adjusts the interval between the conveying surface 22 and the ink ejection surface 28 of the recording head 26, and adjusts the recording position to an appropriate position.

(39) The maintenance unit **16** and the cap unit **17** are arranged below the discharge unit **13**. The maintenance unit **16** performs a wiping operation of wiping the ink ejection surface **28** and cleaning the ink ejection surface **28**. When the wiping operation is executed by the maintenance unit **16**, the conveying unit **12** is arranged at the maintenance position. In this state, the maintenance unit **16** is moved to a position below the recording unit **14**.

(40) The cap unit **17** is moved below the recording unit **14** when capping the ink ejection surface **28** (see FIG. 2) of the recording head **26**. The cap unit **17** then moves upward from below the recording unit **14** to cover the ink ejection surface **28** of the lower surface of the recording head **26** to protect the ink ejection surface **28** from drying.

(41) FIG. 5 is a block diagram showing the configuration of the recording apparatus **10**. As shown in FIG. 5, the control device **80** includes a control portion **81**, a ROM **82**, a RAM **83**, a storage device **84**, and the like. The control device **80** also includes a signal interface that converts a signal to be input or output into a signal having a specified format, a communication device that performs data communication with other devices, and the like. The control device **80** is electrically connected to each component of the recording apparatus **10**, and controls each component of the recording apparatus **10** based on image data input from the outside, and executes image recording processing. In addition, the control device **80** executes various types of data processing, head height adjustment processing, which will be described later, and the like.

(42) The storage device **84** is a computer-readable storage device. The storage device **84** is an example of a specified storage portion according to the present disclosure. The storage device **84** is, for example, a flash memory or a hard disk drive. The storage device **84** stores a specified control program executed by the control portion **81**, various types of data used for various types of processes executed by the control portion **81**, and the like. In addition, the storage device **84** also stores identification information indicating a type of the recording medium **Ct** and the optimum head height **H** in a case where image recording is performed on the recording medium **Ct** in a state of being associated with each other.

(43) Note that the storage portion according to the present disclosure is not limited to the storage device **84** provided in the control device **80**. For example, the storage portion may be an external storage device connected to the control portion **81** via a network by the communication device. In this case, the control portion **81** performs processing for storing data in the external storage device and retrieving data from the external storage device by communicating with the external storage device via the network.

(44) The operation display portion **18** includes an operation portion that receives operations of the user using the recording apparatus **10**. The operation portion is, for example, operation buttons, a touch panel, and the like. For example, the operation portion receives an execution instruction for executing head height adjustment processing, which will be described later, and an input operation for specifying information and data necessary for the processing. That is, the operation portion includes an execution key for inputting the execution instruction, various types of input keys for performing the input operation, a cancel key for interrupting the process, and the like. In addition, the operation display portion **18** includes a display portion for displaying various types of information. The display portion is, for example, a panel display device such as a liquid crystal display unit.

(45) The control portion **81** has a computing device such as a CPU. The control portion **81** is a processor that executes various types of data processing and control by executing a computer program. The control portion **81** performs overall control of each component of the recording apparatus **10** based on image data input from the outside, thereby recording an image corresponding to the image data on the recording surface of the recording medium **Ct**. In addition, the control portion **81** executes head height adjustment processing, which will be described later, according to the procedure of the flowchart shown in FIG. 5.

(46) The ROM **82** is a non-volatile storage device, and the RAM **83** is a computer-readable volatile

storage device. The ROM **82** stores a specified control program, a program for executing head height adjustment processing, which will be described later, and the like. The RAM **83** is used as temporary storage memory for various types of processes executed by the control portion **81**. The RAM **83** temporarily stores the computer program executed by the CPU of the control portion **81** and stores data that is output and referred to while the CPU of the computer portion **81** executes various types of processes. Note that the RAM **83** may be used as a specified storage portion according to the present disclosure.

(47) The control portion **81** includes a plurality of processing portions achieved by executing the computer program. More specifically, the control portion **81** includes a pattern print processing portion **811**, a calculation processing portion **812**, a determination processing portion **813**, a storage processing portion **814** (an example of a storage processing portion according to the present disclosure), a display processing portion **815** (an example of an output processing portion according to the present disclosure), a position adjustment processing portion **816**, and the like. Each of these processing portions is not limited to being achieved by the control portion **81**, and may be achieved by, for example, a plurality of processors. In addition, each processing portion is not limited to being a processing portion achieved by the control portion **81**, and may be configured by, for example, an electronic circuit capable of achieving the function of each processing portion.

(48) The pattern print processing portion **811** performs processing for causing the recording head **26** to record a specified test pattern on the recording surface of the recording medium Ct. For example, in a case where an image is recorded on the recording medium Ct whose thickness information is not registered in the recording apparatus **10**, the user operates the operation display portion **18** of the recording apparatus **10** to input the identification information of the recording medium Ct (for example, the name of the medium). After that, the user presses and operates an execution key for executing head height adjustment processing, which will be described later, to cause the recording apparatus **10** to start the head height adjustment processing. In this case, the control portion **81** drives and controls the raising/lowering mechanism **15** to move the conveying unit **12** so that the head height H becomes a predetermined height H1. Here, the specified height H1 may be a position that maintains a sufficient interval T that prevents the recording surface of the recording medium Ct from coming into contact with the ink ejection surface **28** even when the recording medium Ct with an unknown thickness is conveyed on the conveying surface **22**. After that, the pattern print processing portion **811** performs processing for recording the test pattern on the recording medium Ct set in the recording apparatus **10**.

(49) Here, the test pattern is a linear image elongated in the conveying direction D1 of the recording medium Ct. Note that the test pattern may have any shape, such as a rectangular shape, a dot shape, or the like, as long as the shape is such that the control portion **81** can determine the recording position of the test pattern on the image data captured by the imaging portion **19**.

(50) In this embodiment, the pattern print processing portion **811** executes a first print process for printing a first test pattern TP1 and a second print process for printing a second test pattern TP2. The first print process is a process in which the recording head **26** is moved in the first direction D21 of the scanning direction D2, and ink droplets are ejected toward the recording surface of the recording medium Ct during the movement to print the first test pattern TP1 (see FIG. 6) on the recording surface. In addition, the second print process is a process in which the recording head **26** is moved in the second direction D22 of the scanning direction D2, and ink droplets are ejected toward the recording surface of the recording medium Ct during the movement to print the second test pattern TP2 (see FIG. 6) on the recording surface. The second direction D22 is a direction opposite to the first direction D21 in the scanning direction D2. The position where the second test pattern TP2 is printed is an example of the reference position according to the present disclosure.

(51) In a case where the recording head **26** performs forward movement (movement in the first direction D21) and backward movement (movement in the second direction D22) along the scanning direction D2, in each movement process, the pattern print processing portion **811** prints

each test pattern TP1, TP2 on the recording surface. More specifically, when the recording head 26 reaches a predetermined specified position P10, the pattern print processing portion 811 prints each of the test patterns TP1 and TP2 on the recording surface by ejecting ink droplets toward the recording surface.

(52) Here, as shown in FIG. 6, the specified position P10 is a predetermined position on the recording surface of the recording medium Ct and a specific position in the scanning direction D2. In this embodiment, the specified position P10 is a central position in the scanning direction D2 on the recording surface of the recording medium Ct. That is, the specified position P10 is the center position in the width direction of the recording medium Ct.

(53) When the recording head 26 reaches the specified position P10 while the recording head 26 is moving in the first direction D21, ink droplets are ejected toward the recording surface to print the test pattern TP1 on the recording surface. In addition, when the recording head 26 reaches the specified position P10 while the recording head 26 is moving in the second direction D22, ink droplets are ejected toward the recording surface to print the test pattern TP2 on the recording surface. Note that the control portion 81 can determine that the recording head 26 has reached the specified position P10 based on a sense signal from a rotary encoder or the like provided in the motor 25.

(54) Ink droplets ejected in the process of moving in the first direction D21 are ejected in a direction inclined in the movement direction D21 of the recording head 26 and land on the recording surface. In addition, ink droplets ejected in the process of moving in the second direction D22 are ejected in a direction inclined in the movement direction D22 of the recording head 26 and land on the recording surface. Therefore, the test patterns TP1 and TP2 are recorded at positions separated by a predetermined interval DS (hereinafter referred to as “pattern interval DS”) in the scanning direction D2 so as to sandwich the specified position P10 therebetween.

(55) When the test patterns are printed on the recording surface of the recording medium Ct, the imaging portion 19 reads an image including the test patterns TP1 and TP2 (hereinafter referred to as a “pattern image”).

(56) The calculation processing portion 812 performs a process of calculating the pattern interval DS between the first test pattern TP1 and the second test pattern TP2 included in the pattern image based on the pattern image read by the imaging portion 19. The pattern interval DS is a distance between the respective center positions of the test patterns TP1 and TP2. The pattern interval DS can be calculated based on the number of pixels between each test pattern in the pattern image and the reading resolution of the imaging portion 19 (for example, the number of pixels per inch).

(57) The determination processing portion 813 performs a process for determining the head height H of the recording head 26 (height position of the recording head 26) according to the recording medium Ct based on the pattern interval DS. The head height H is the interval of the interval T such that the separation distance t1 (see FIGS. 3 and 4A) between the ink ejection surface 28 and the recording surface of the recording medium Ct becomes the design interval t0 (see FIG. 4A) in a case where the recording medium Ct on which the test pattern is printed is conveyed on the conveying surface 22.

(58) Here, after adjusting the head height H so that the separation distance t1 becomes the design interval t0, the test patterns TP1 and TP2 are printed on a reference recording medium having a known thickness, and the interval between the patterns is taken to be DS0. In this case, the interval DS0, the pattern interval DS, the design interval t0, and the separation distance t1 (the distance between the ink ejection surface 28 and the recording surface) are in a proportional relationship of DS0:DS=t0:t1. Here, the interval DS0 is calculated in advance by the control portion 81 based on the pattern image including the test patterns TP1 and TP2 printed on the reference recording medium.

(59) From the proportional relationship, the separation distance t1 can be expressed by the following equation (1).

$$t1=(DS.Math.t0)/DS0 \quad (1)$$

(60) Here, when the predetermined specified height H1 (the distance of the interval T) is used in a case where the head position adjustment process is performed, a thickness k of the recording medium Ct can be expressed by the following equation (2).

$$k=H1-t1=H1-(DS.Math.t0)/DS0 \quad (2)$$

(61) Therefore, the optimum value of the head height H of the recording head **26** according to the recording medium Ct is derived from a relational expression indicated by equation (3) below.

$$H=k+t0=H1-(DS.Math.t0)/DS0+t0 \quad (3)$$

(62) In this embodiment, the relational expression and information for deriving the relational expression are stored in a storage device **84**, and the determination processing portion **813** calculates the appropriate head height H using the relational expression and information in the storage device **84** and the pattern interval DS calculated by the calculation processing portion **812**.

(63) The storage processing portion **814** performs a process of associating the identification information (for example, medium name) of the recording medium Ct input by the user from the operation display portion **18** with the head height H determined by the determination processing portion **813**, and storing the result in the storage device **84**.

(64) The display processing portion **815** performs a process for outputting information related to the head height H determined by the determination processing portion **813**. Information related to the head height H may be a numerical value of the head height H itself, or may indicate whether the head height H is higher or lower than a setting value of the head height H (optimum setting value). An output destination is, for example, an external device connected via a network to the recording apparatus **10** or is a display portion of the operation display portion **18**. The external device is, for example, a terminal device or an information processing device possessed by a user who uses the recording apparatus **10**. In addition, the display processing portion **815** reads the identification information and the head height H from the storage device **84** and performs processing for displaying the identification information and the head height H on the display portion of the operation display portion **18**.

(65) The position adjustment processing portion **816** performs processing that causes the raising/lowering mechanism **15** to adjust the relative position of the recording head **26** with respect to the conveying surface **22** so that the recording head **26** is arranged at the head height H determined by the determination processing portion **813**.

(66) In a conventional recording apparatus, in order to determine the distance to a recording medium of unknown thickness, not only is a test pattern printed on the recording medium, but it is necessary to print a test pattern on a recording medium of known thickness in advance, and thus the printing work is complicated.

(67) However, since the recording apparatus **10** according to the present embodiment is configured as described above, it is possible to appropriately determine the height position of the recording head according to the recording medium such as fabric or paper. That is, with the recording apparatus **10**, even in a case where the image recording process is performed on a recording medium Ct of unknown thickness, it is possible to determine a setting value (optimal setting value) of the head height H of the recording head **26** that can ensure image quality equal to or higher than a reference level. In addition, it is possible to automatically adjust the position of the recording head **26** in the height direction to an appropriate position based on the determined setting value.

(68) A head height adjustment process and a head height adjustment method executed by the control portion **81** will be described below with reference to the flowchart of FIG. 7. Here, FIG. 7 is a flowchart showing an example of a procedure of the head height adjustment process executed by the control portion **81** of the recording apparatus **10**. In FIG. 7, S11, S12, and so on, represent the numbers of the processing procedures (steps).

(69) Note that in the following description, it is presumed that the operation mode of the recording apparatus **10** has been changed from a normal print mode to an adjustment mode for adjusting the

head height H of the recording head **26** by the user operating the operation display portion **18** before recording an image on the recording medium Ct whose thickness information is not registered in the recording apparatus **10**.

(70) First, in step **S11**, the control portion **81** determines whether or not information of a new recording medium Ct has been input in the adjustment mode. For example, the user inputs the identification information (such as the name of the recording medium) from the operation display portion **18** as the information of the recording medium Ct whose thickness information is not registered in the recording apparatus **10**. When the identification information is input, the control portion **81** registers the identification information in the recording apparatus **10**. More specifically, the identification information is registered as information of the new recording medium Ct in the registration data of the recording medium Ct stored in the storage device **84**.

(71) After that, in step **S12**, the control portion **81** determines whether or not the setting value of the head height H (height setting value) corresponding to the new recording medium Ct has been input. For example, the user can input an arbitrary height setting value from the operation display portion **18** as a temporary setting value of the head height H corresponding to the new recording medium Ct. When the height setting value is input, the control portion **81** associates the height setting value with the identification information registered in step **S11**, and registers the height setting value in the registration data.

(72) In the next step **S13**, the control portion **81** drives and controls the raising/lowering mechanism **15** to raise or lower the conveying unit **12** so that the head height H of the recording head **26** is at the position indicated by the height setting value input in step **S12**. That is, the control portion **81** drives and controls the raising/lowering mechanism **15** to adjust the interval of the interval T so that the head height H becomes the height setting value. After that, the process proceeds to step **S15**.

(73) On the other hand, in a case where the height setting value is not input in step **S12**, the control portion **81** reads the specified height H1 stored in advance in the storage device **84** or the like, and drives and controls the raising/lowering mechanism **15** to raise or lower the conveying unit **12** so that the head height H of the recording head **26** becomes the specified height H1 (**S14**). After that, the process proceeds to step **S15**.

(74) In step **S15**, it is determined whether or not an adjustment instruction for adjusting the head height H has been input. When the user inputs an execution instruction by operating the operation display portion **18**, the control portion **81** receives an input signal of the execution instruction and determines that the adjustment instruction has been input.

(75) When the adjustment instruction is input, the control portion **81** conveys the recording medium Ct set on the upper surface of the feed unit **11** to the recording position below the recording head **26** (**S16**). After that, the control portion **81** performs processing (**S17** to **S20**) for printing the test patterns on the recording surface of the recording medium Ct that is paused at the recording position. Note that this process is executed by the pattern print processing portion **811** (see FIG. 5).

(76) More specifically, first, the control portion **81** moves the recording head **26** in the first direction D21 from a specified standby position (**S17**). When the recording head **26** reaches the specified position P10, the control portion **81** ejects ink droplets onto the recording surface of the recording medium Ct to print the first test pattern TP1 on the recording surface (**S18**). In addition, the control portion **81** moves the recording head **26** in the second direction D22 by returning from the end portion on the first direction D21 side (**S19**). When the recording head **26** reaches the specified position P10 again, the control portion **81** causes the recording head **26** to eject ink droplets toward the recording surface of the recording medium Ct to print the second test pattern TP2 on the recording surface (**S20**). Note that steps **S17** to **S20** are an example of the test pattern printing step according to the present disclosure.

(77) After the test patterns TP1 and TP2 are printed, the control portion **81** stops the operation of

the recording head **26** and performs processing to discharge the recording medium Ct (**S21**). In addition, the control portion **81** causes the imaging portion **19** to read the pattern image including the test patterns TP1 and TP2 in the process of discharging the recording medium Ct (**S22**). After that, the control portion **81** calculates the interval DS (pattern interval DS) between the test pattern TP1 and the test pattern TP2 based on the pattern image (**S23**). This processing is processing executed by the calculation processing portion **812** (see FIG. 5). Note that step **S22** is an example of the imaging step according to the present disclosure. In addition, step **S23** is an example of the interval calculation step according to the present disclosure.

(78) When the pattern interval DS is calculated, the control portion **81**, in step **S24**, determines whether or not the pattern interval DS is within a predetermined allowable range. Here, the allowable range is a numerical range obtained by adding a specified margin (allowable amount) to the interval DS0 when the interval between the ink ejection surface **28** and the recording surface of the recording medium Ct is the design interval t0. For example, in a case where the margin is taken to be α , the allowable range is from “DS0- α ” to “DS0+ α ”.

(79) In step **S24**, in a case where the pattern interval DS is determined to be within the allowable range, that is, in a case where the pattern interval DS is determined to be equal to or greater than a lower limit value of the allowable range and equal to or less than an upper limit value of the allowable range, the control portion **81** registers the current head height H in the registration data (**S28**). After that, this series of processing ends.

(80) On the other hand, in a case where it is determined in step **S24** that the pattern interval DS is outside the allowable range, that is, in a case where it is determined that the pattern interval DS is smaller than the lower limit of the allowable range or larger than the upper limit of the allowable range, the control portion **81** proceeds to the next step **S25**, and performs processing (head height determination processing) for determining the optimum setting value of the head height H corresponding to the recording medium Ct. This processing is processing executed by the determination processing portion **813** (see FIG. 5). The processing of step **S25** is performed using the relational expression indicated as equation (3) above. Note that step **S25** is an example of the height position determination step according to the present disclosure.

(81) When the head height H is determined, the control portion **81**, in step **S26**, performs processing to output the setting value of the determined head height H to the operation display portion **18** and display the setting value on the display portion of the operation display portion **18**. At this time, the control portion **81** reads the identification information of the recording medium Ct from the storage device **84**, associates the identification information with the setting value of the head height H, and displays the result on the display portion. This processing is processing executed by the display processing portion **815** (see FIG. 5).

(82) After that, in step **S27**, the control portion **81** drives and controls the raising/lowering mechanism **15** to raise or lower the conveying unit **12** so that the head height H of the recording head **26** is at the position indicated by the height setting value determined in step **S25**. That is, the control portion **81** drives and controls the raising/lowering mechanism **15** to change the interval of the interval T so that the head height H becomes the height setting value. This processing is processing executed by the position adjustment processing portion **816** (see FIG. 5). After that, the setting value of the head height H after the change is registered in the registration data (**S28**), and this series of processing ends. Note that the processing of step **S28** is processing executed by the storage processing portion **814** (see FIG. 5).

(83) As described above, according to the present embodiment, the test patterns TP1 and TP2 are printed on the recording surface of the recording medium Ct by the control portion **81** (**S17-S20**), the pattern image including the test patterns TP1 and TP2 is read by the imaging portion **19**, after that, the interval DS (pattern interval DS) between the test patterns TP1 and TP2 is calculated (**S23**), then, the optimum setting value for the head height H is determined based on the pattern interval DS (**S25**). Thus, even in a case where the image recording process is performed on a

recording medium Ct whose thickness is unknown, in the adjustment mode before the image recording process, it is possible to determine a setting value (optimum setting value) of the head height H of the recording head **26** that can ensure image quality equal to or higher than a reference level.

(84) In addition, the control portion **81** automatically adjusts the position of the recording head **26** in the height direction to an appropriate position so that the height position indicated by the determined optimum setting value is obtained. Thus, the user can be saved the trouble of manually adjusting the head height H.

(85) In addition, since the control portion **81** displays the optimum setting value for the head height H on the display portion of the operation display portion **18**, the user is able to know the optimum setting value for the head height H corresponding to the recording medium Ct.

(86) Note that in the embodiment described above, an example in which the position in the height direction of the recording head **26** is automatically changed by the raising/lowering mechanism **15** has been described; however, for example, in a case where the raising/lowering mechanism **15** does not have an automatic raising/lowering function, it is possible, for example, for the user to confirm the setting value displayed on the operation display portion **18** and input the numerical value as the head height H of the recording head **26** from the operation display portion **18**. In this case, the control portion **81** drives and controls the raising/lowering mechanism **15** based on the input setting value to raise or lower the conveying unit **12** so that the head height H of the recording head **26** is at the position indicated by the input setting value.

(87) In addition, in the embodiment described above, a processing example of recording one first test pattern TP1 and one second test pattern TP2 on the recording surface of the recording medium Ct has been described; however, the present disclosure is not limited to this processing example. For example, sets of two, the first test pattern TP1 and the second test pattern TP2, may be set, and a plurality of such sets may be recorded on the recording surface. It is preferable that the plurality of sets of test patterns be recorded at each of a plurality of positions separated in the scanning direction D2 on the recording surface of the recording medium Ct. In this case, the control portion **81** calculates the pattern intervals DS of each of the plurality of sets of test patterns from the pattern image read by the imaging portion **19**, and calculates the average value of the pattern intervals DS. The control portion **81** obtains the optimum head height H using the average value and the relational expression shown in equation (3) above. With this configuration, the optimum value of the head height H can be determined with higher accuracy.

(88) In addition, in the embodiment described above, a processing example in which two test patterns TP1 and TP2 are printed on the recording surface of the recording medium Ct has been described; however, processing is not limited to processing such as described in the present disclosure. For example, a processing example of printing only the first test pattern TP1 on the recording surface of the recording medium Ct can be considered. In this case, when a position (specified position) on the recording surface corresponding to the specified position P10 is taken to be a reference position, the control portion **81** calculates an interval DS from the reference position to the first test pattern TP1, and obtains the optimum head height H using this interval DS and the relational expression shown in the equation (3) above. Even in such a processing example, the optimum value of the head height H can be determined.

(89) It is to be understood that the embodiments herein are illustrative and not restrictive, since the scope of the disclosure is defined by the appended claims rather than by the description preceding them, and all changes that fall within metes and bounds of the claims, or equivalence of such metes and bounds thereof are therefore intended to be embraced by the claims.

Claims

1. An inkjet recording apparatus, comprising: a recording head configured to eject ink onto a recording medium while moving in a predetermined scanning direction; a pattern print processing portion configured to cause the recording head to record a specified test pattern on a recording surface of the recording medium; an imaging portion configured to capture an image of the recording surface including the test pattern; a calculation processing portion configured to calculate an interval between the test pattern and a reference position on the recording surface based on the image of the recording surface; and a determination processing portion configured to determine a height position of the recording head according to the recording medium based on the interval; and further comprising: one of either an output processing portion, or a head position adjustment portion and a position adjustment processing portion; wherein the output processing portion is configured to output information related to the height position determined by the determination processing portion; the head position adjustment portion is configured to be able to adjust the height position by changing a relative position of the recording head with respect to a support surface of the recording medium; and the position adjustment processing portion is configured to cause the head position adjustment portion to change the relative position of the recording head based on the height position determined by the determination processing portion, wherein the pattern print processing portion executes a first printing process of moving the recording head in a first direction of the scanning direction to print a first test pattern at a specified position on the recording surface, and a second printing process of moving the recording head in a second direction opposite to the first direction in the scanning direction to print a second test pattern as the reference position at a specified position on the recording surface; the calculation processing portion calculates the interval between the first test pattern and the second test pattern based on the image of the recording surface; the pattern print processing portion records the test pattern at each of a plurality of positions separated in the scanning direction on the recording surface; the calculation processing portion calculates an average value of the plurality of intervals corresponding to each of the plurality of test patterns included in the image of the recording surface; and the determination processing portion determines the height position based on the average value.
2. The inkjet recording apparatus according to claim 1, wherein the determination processing portion calculates the height position corresponding to the interval using a relational expression indicating a relationship between the interval and the height position.
3. The inkjet recording apparatus according to claim 1, further comprising: a storage processing portion configured to associate identification information of the recording medium and the height position determined by the determination processing portion and store the identification information and the height position in a specified storage portion; wherein the output processing portion reads the identification information and the height position from the storage portion and displays the identification information and the height position on a specified display portion.
4. A head position determining method comprising: a test pattern printing step of recording a specified test pattern on a recording surface of a recording medium by ejecting ink from a recording head while moving the recording head in a predetermined scanning direction, executing a first printing process of moving the recording head in a first direction of the scanning direction to print a first test pattern at a specified position on the recording surface, and a second printing process of moving the recording head in a second direction opposite to the first direction in the scanning direction to print a second test pattern as the reference position at a specified position on the recording surface, recording the test patterns at the specified position and the reference position separated in the scanning direction on the recording surface; an image capturing step of capturing an image of the recording surface including the test pattern; an interval calculation step of calculating an interval between the first test pattern and the second test pattern based on the image of the recording surface, and calculating an average value of a plurality of intervals corresponding to each of the plurality of test patterns included in the image of the recording surface; and a height

position determination step of determining a height position of the recording head according to the recording medium based on the interval and the average value.
